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博士 · 副研究员

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研究方向

以**大模型推理与智能体**为核心研究方向，聚焦情境感知推理、自主交互学习与持续适应，探索其在时序数据挖掘、科技文献与科学工具挖掘中的理论方法与应用。

核心问题：在信息不完整、环境会变化、反馈有成本的条件下，智能体如何通过推理与交互可靠地解决问题?

- **大模型智能体与自主学习**。研究环境交互式智能体强化学习、工具增强推理、多智能体协同，以及基于情境、知识与记忆的持续能力进化。
- **时序数据挖掘**。研究多模态情境表征、慢思考推理、不确定性感知预测与自主智能体交互，关注电力负荷、能源发电与气象等垂直领域的时序预测应用。
- **科学智能：科技文献与科学工具挖掘**。研究科技文献检索、多模态文献理解、文本与图表中的知识抽取、跨文献知识整合、文献证据驱动推理，以及科学工具挖掘。

工作与科研经历

副研究员 2023.06 – 至今

中国科学技术大学计算机科学与技术学院，合肥

认知智能全国重点实验室。

研究实习生 2019.06 – 2019.10

腾讯平台与内容事业群（PCG）数据服务中心（DSC），深圳

工程实习生 2018.07 – 2018.09

科大讯飞核心技术研发平台大数据实验室，合肥

教育经历

计算机科学与技术，博士 2017.09 – 2023.06

中国科学技术大学

导师：刘淇教授。

市场营销，管理学学士 2013.09 – 2017.07

合肥工业大学

研究工作定位

以大模型智能体与自主学习为通用方法基础，面向科技文献、科学工具和时序等结构化数据，围绕语义理解、认知推理、自主交互与持续进化开展研究。应用场景包括科学研究、个性化智能与预测智能。

论文与研究成果

完整收录个人主页中展示的 77 条记录，包括演示论文、Workshop 成果及预印本。* 通讯作者；+ 同等贡献。

2027 年

[1] **Mingyue Cheng**, Xiaoyu Tao, Ze Guo, Qi Liu*, Bokai Pan, Shangtong Ye, Shijin Wang, Enhong Chen. One Forecast, One World: Adaptive Evidence Reasoning for Context-Aware Time Series Forecasting (AdaCast). *ACM WSDM 2027* 已录用.

[2] Daoyu Wang, **Mingyue Cheng***, Shuo Yu, Zirui Liu, Ze Guo, Xin Li, Qi Liu. PaperArena: An Evaluation Benchmark for Tool-Augmented Agentic Reasoning on Scientific Literature. *ACM WSDM 2027* 已录用. [Code]

[3] Panjing He, **Mingyue Cheng***, Yucong Luo, Li Li, Xiaohan Zhang. SheetCompass: Hierarchical Relation Graphs for Agentic Spreadsheet Reasoning. *ACM WSDM 2027* 已录用. [Code]

2026 年

[4] **Mingyue Cheng**, Yucong Luo, Jie Ouyang, Qi Liu*, Huijie Liu, Li Li, Shuo Yu, Bohou Zhang, Jiawei Cao, Jie Ma, Daoyu Wang, Enhong Chen. A Survey on Knowledge-Oriented Retrieval-Augmented Generation. *ACM Transactions on Information Systems (ACM TOIS)* 已录用. [Code]

[5] **Mingyue Cheng**, Zhiding Liu, Xiaoyu Tao, Qi Liu*, Jintao Zhang, Tingyue Pan, Shilong Zhang, Panjing He, Xiaohan Zhang, Daoyu Wang, Jiahao Wang, Enhong Chen. A Comprehensive Survey of Time Series Forecasting: Concepts, Challenges, and Future Directions. *IEEE Transactions on Knowledge and Data Engineering (IEEE TKDE)* 已录用. [Code]

[6] **Mingyue Cheng**, Qingyang Mao, Qi Liu*, Yitong Zhou, Yupeng Li, Jiahao Wang, Jiaying Lin, Jiawei Cao, Enhong Chen. A Survey on Table Mining with Large Language Models: Challenges, Advancements and Prospects. *ACM Computing Surveys (ACM CSUR)* 已录用. [Code]

[7] Tingyue Pan, **Mingyue Cheng***, Shilong Zhang, Zhiding Liu, Xiaoyu Tao, Yucong Luo, Jintao Zhang, Qi Liu. OneCast: Structured Decomposition and Modular Generation for Cross-Domain Time Series Forecasting. *ACM Transactions on Knowledge Discovery from Data (ACM TKDD)* 已录用.

[8] **Mingyue Cheng**, Daoyu Wang, Qi Liu*, Shuo Yu, Xiaoyu Tao, Yuqian Wang, Chengzhong Chu, Yu Duan, Mingkang Long, Enhong Chen. Mind2Report: A Cognitive Deep Research Agent for Expert-Level Commercial Report Synthesis. *ACM CIKM 2026* 已录用. [Code]

[9] Yitong Zhou, Yucong Luo, **Mingyue Cheng***, Jiahao Wang, Daoyu Wang, Tingyue Pan, Jintao Zhang, Qi Liu, Enhong Chen. Time Series Forecasting as Reasoning: A Slow-Thinking Approach with Reinforced LLMs. *ACM CIKM 2026* 已录用. [Code]

[10] Xiaohan Zhang, Tian Gao, **Mingyue Cheng***, Bokai Pan, Ze Guo, Yaguo Liu, Xiaoyu Tao, Qi Liu. AlphaCast: A Human Wisdom-LLM Intelligence Co-Reasoning Framework for Interactive Time Series Forecasting. *ACM CIKM 2026* 已录用. [Code]

[11] **Mingyue Cheng**, Xiaoyu Tao, Huajian Zhang, Qi Liu*, Enhong Chen. InstructTime++: Time Series Classification with Multimodal Language Modeling via Implicit Feature Enhancement. *ACM Transactions on Intelligent Systems and Technology (ACM TIST)* 已录用. [Code]

[12] Yitong Zhou, **Mingyue Cheng***, Qingyang Mao, Yucong Luo, Qi Liu, Yupeng Li, Xiaohan Zhang, Deguang Liu, Xin Li, Enhong Chen. Benchmarking Multimodal LLMs on Recognition and Understanding over Chemical Tables. *KDD 2026* 已录用. [Code]

[13] Xiaoyu Tao, **Mingyue Cheng***, Ze Guo, Shuo Yu, Yaguo Liu, Qi Liu, Shijin Wang. MemCast: Memory-Driven Time Series Forecasting with Experience-Conditioned Reasoning. *ICML 2026* 已录用.

[14] **Mingyue Cheng**, Yaguo Liu, Daoyu Wang, Xiaoyu Tao, Qi Liu*. CoGenCast: A Coupled Autoregressive-Flow Generative Framework for Time Series Forecasting. *ICML 2026* 已录用.

[15] Xiaoyu Tao, Shilong Zhang, **Mingyue Cheng***, Daoyu Wang, Tingyue Pan, Bokai Pan, Changqing Zhang, Shijin Wang. From Values to Tokens: An LLM-Driven Framework for Context-aware Time Series Forecasting via Symbolic Discretization. *IJCAI 2026* 已录用.

[16] Bohou Zhang, Xiaoyu Tao, **Mingyue Cheng***, Huijie Liu, Qi Liu. ScholarSum: Student-Teacher Abstractive Summarization via Knowledge Graph Reasoning and Reflective Refinement. *IJCAI 2026* 已录用.

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[18] Huajian Zhang, **Mingyue Cheng***, Yucong Luo, Xiaoyu Tao. STaR: Towards Cognitive Table Reasoning via Slow-Thinking Large Language Models. *ACM WWW 2026*. [Code]

[19] Zhiding Liu, Ben Chen, **Mingyue Cheng***, Enhong Chen, Li Li, Chenyi Lei*, Wenwu Ou, Han Li and Kun Gai. Towards Context-aware Reasoning-enhanced Generative Searching in E-commerce. *ACM WWW 2026*.

[20] Shuo Yu, **Mingyue Cheng***, Daoyu Wang, Qi Liu, Zirui Liu, Ze Guo, Xiaoyu Tao. MemWeaver: A Hierarchical Memory from Textual Interactive Behaviors for Personalized Generation. *ACM WWW 2026*.

[21] **Mingyue Cheng**, Jiahao Wang, Daoyu Wang, Xiaoyu Tao, Qi Liu*, Enhong Chen. Can Slow-Thinking LLMs Reason Over Time? Empirical Studies in Time Series Forecasting. *ACM WSDM 2026*: 99–110, Feb 2026. [Code]

[22] **Mingyue Cheng**, Xiaoyu Tao, Zhiding Liu, Qi Liu*, Hao Zhang, Rujiao Zhang and Enhong Chen. TimeMAE: Self-Supervised Representations of Time Series with Decoupled Masked Autoencoders. *ACM WSDM 2026*: 498–508, Feb 2026. [Code]

[23] Chuang Jiang, **Mingyue Cheng***, Xiaoyu Tao, Qingyang Mao, Jie Ouyang and Qi Liu. TableMind: An Autonomous Programmatic Agent for Tool-Augmented Table Reasoning. *ACM WSDM 2026*: 260–270, Feb 2026. [Code]

[24] Qingchuan Li, **Mingyue Cheng***, Zirui Liu, Daoyu Wang, Yuting Zeng, Tongxuan Liu. From Hypothesis to Premises: LLM-based Backward Logical Reasoning with Selective Symbolic Translation. *AAAI 2026*: 31671–31679, Mar 2026.

[25] Yupeng Li, **Mingyue Cheng***, Yucong Luo, Yitong Zhou, Qingyang Mao, Shijin Wang. BLADE: A Behavior-Level Data Augmentation Framework with Dual Fusion Modeling for Multi-Behavior Sequential Recommendation. *AAAI 2026*: 15189–15197, Mar 2026.

[26] Tingyue Pan, Jie Ouyang, **Mingyue Cheng**, Qingchuan Li, Zirui Liu, Daoyu Wang, Mingfan Pan, Shuo Yu, Qi Liu. PaperScout: An Autonomous Agent for Academic Paper Search with Process-Aware Sequence-Level Policy Optimization. *Findings of EMNLP 2026* 已录用.

[27] Xiaoyu Tao, **Mingyue Cheng**, Ze Guo, Bokai Pan, Qi Liu, Shijin Wang, Enhong Chen. CastClaw: A Human-in-the-Loop Autonomous Agent for Interactive Time Series Forecasting. *IEEE ICDM 2026 Demo Track* 已录用. [Code]

[28] Daoyu Wang, **Mingyue Cheng**, Qingchuan Li, Shuo Yu, Jie Ouyang, Qi Liu, Enhong Chen. AgentWorld: A Data Factory for Training LLM Agents from Multi-Turn Interactions. *IEEE ICDM 2026 Demo Track* 已录用. [Code]

[29] **Mingyue Cheng**, Shuo Yu, Daoyu Wang, Qingchuan Li, Xiaoyu Tao, Jie Ouyang, Yucong Luo, Yitong Zhou, Qi Liu*, Enhong Chen. Agent-R1: A Unified and Modular Framework for Agentic Reinforcement Learning. *ACM CIKM 2026 Demo Track* 已录用.

[30] **Mingyue Cheng**, Shuo Yu, Daoyu Wang, Qingchuan Li, Xiaoyu Tao, Qingyang Mao, Yitong Zhou, Qi Liu*. TabClaw: An Interactive and Self-Evolving Agent for Spreadsheet Manipulation and Table Reasoning. *ACM CIKM 2026 Demo Track* 已录用.

2025 年

[31] **Mingyue Cheng**, Jiqian Yang, Tingyue Pan, Qi Liu*, Zhi Li. ConvTimeNet: A Deep Hierarchical Fully Convolutional Model for Multivariate Time Series Analysis. *ACM WWW 2025*, Sydney, 2025. — Included in *sktime*. [Code]

[32] Xiaoyu Tao, Tingyue Pan, **Mingyue Cheng***, Yucong Luo, Qi Liu, Enhong Chen. Hierarchical Multimodal LLMs with Semantic Space Alignment for Enhanced Time Series Classification. (*ACM Transactions on Intelligent Systems and Technology (ACM TIST)*). [Code]

[33] **Mingyue Cheng**, Yiheng Chen, Qi Liu*, Zhiding Liu, Yucong Luo, Enhong Chen. InstrucTime: Advancing Time Series Classification with Multimodal Language Modeling. *ACM WSDM 2025* (**Best of WSDM**): 792–800, Hannover, 2025. [Code] [Poster]

[34] **Mingyue Cheng**, Xiaoyu Tao, Qi Liu*, Hao Zhang, Yiheng Chen, Defu Lian. Cross-Domain Pre-training with Language Models for Transferable Time Series Representations. *ACM WSDM 2025*: 175–183, Hannover, 2025. [Code] [Poster]

[35] Daoyu Wang, **Mingyue Cheng***, Zhiding Liu, Qi Liu. TimeDART: A Diffusion Autoregressive Transformer for Self-supervised Time Series Representation. *ICML 2025*, Vancouver, PMLR 267, 2025. [Code]

[36] Jie Ouyang, Tingyue Pan, **Mingyue Cheng***, Ruiran Yan, Yucong Luo, Jiaying Lin, Qi Liu. HoH: A Dynamic Benchmark for Evaluating the Impact of Outdated Information on RAG. *ACL 2025*: 6036–6063, Vienna, 2025. [Code]

[37] **Mingyue Cheng**, Jintao Zhang, Zhiding Liu, Chunli Liu*. A Hybrid Multi-Factor Framework for Dynamic Intraoperative Hypotension Prediction. *IJCAI 2025*: 4923–4931, Montreal, 2025.

[38] Jintao Zhang, **Mingyue Cheng***, Xiaoyu Tao, Zhiding Liu, Daoyu Wang. Conditional Denoising Meets Polynomial Modeling: A Flexible Decoupled Framework for Time Series Forecasting. *IJCAI 2025*: 6993–7001, Montreal, 2025. [Code]

[39] Yitong Zhou, **Mingyue Cheng***, Qingyang Mao, Feiyang Xu, Xin Li. Enhancing Table Recognition with Vision LLMs: A Benchmark and Neighbor-Guided Toolchain Reasoner. *IJCAI 2025*: 2503–2511, Montreal, 2025. [Code]

[40] Hao Zhang, **Mingyue Cheng***, Qi Liu, Zhiding Liu, Linbo Zhu, Yu Su. Towards Automatic Sampling of User Behaviors for Sequential Recommender Systems. *IJCAI 2025*: 3624–3632, Montreal, 2025. [Code]

[41] Jiahao Wang, **Mingyue Cheng***, Qingyang Mao, Qi Liu, Feiyang Xu, Xin Li, Enhong Chen. TableTime: Reformulating Time Series Classification as Zero-Shot Table Understanding via LLMs. *ACM CIKM 2025*. [Code]

[42] Shuo Yu, **Mingyue Cheng***, Jiqian Yang, Jie Ouyang, et al. Multi-Source Knowledge Pruning for Retrieval-Augmented Generation: A Benchmark and Empirical Study. *ACM CIKM 2025*.

2024 年

[43] Jie Ouyang, Yucong Luo, **Mingyue Cheng***, Shuo Yu, Daoyu Wang, Qi Liu, Enhong Chen. Revisiting the Solution of Meta KDD Cup 2024: CRAG. (KDD Cup Workshop, **2nd Place in Task 2 & Task 3**). [Slides] [Poster]

[44] **Mingyue Cheng**, Qi Liu*, Wenyu Zhang, Zhiding Liu, Hongke Zhao, Enhong Chen. A General Tail Item Representation Enhancement Framework for Sequential Recommender Systems. *Frontiers of Computer Science (FCS)*, 2024, 18(6): 1–12. [Code]

[45] Rujiao Zhang, Hao Zhang, Yucong Luo, Zhiding Liu, **Mingyue Cheng**, Qi Liu, Enhong Chen*. Learning the Dynamics in Sequential Recommendation by Exploiting Real-time Information. *ACM CIKM 2024*: 4288–4292, Oct 2024.

[46] Jie Wang, Fajie Yuan, **Mingyue Cheng**, et al. TransRec: Learning Transferable Recommendation from Mixture-of-Modality Feedback. *APWeb-WAIM 2024*: 193–208, Aug 2024.

[47] Zhiding Liu, Jiqian Yang, **Mingyue Cheng***, Yucong Luo, Zhi Li. Generative Pretrained Hierarchical Transformer for Time Series Forecasting. *ACM SIGKDD 2024*: 2003–2013, Barcelona, 2024.

[48] Yucong Luo, **Mingyue Cheng**, Hao Zhang, Junyu Lu, Enhong Chen*. Unlocking the Potential of Large Language Models for Explainable Recommendations. (DASFAA 2024). [Code]

[49] Junzhe Jiang, Shang Qu, **Mingyue Cheng***, Qi Liu, et al. Reformulating Sequential Recommendation: Learning Dynamic User Interest with Content-enriched Language Modeling. (DASFAA 2024). [Code]

[50] **Mingyue Cheng**, Hao Zhang, Qi Liu*, Fajie Yuan, Zhi Li, Zhenya Huang, Enhong Chen, Longfei Li, Jun Zhou. Empowering Sequential Recommender Systems from Mixture of Collaborative Signals and Semantic Relatedness. (DASFAA 2024). [Code]

[51] Hao Zhang, **Mingyue Cheng***, Qi Liu, Yucong Luo, Rui Li, Enhong Chen. Learning Recommender Systems with Soft Target: A Decoupled Perspective. *DASFAA 2024*, LNCS 14852, Gifu, Japan, 2024. [Code]

[52] **Mingyue Cheng**, Hao Zhang, Jiqian Yang, Qi Liu*, Li Li, Xin Huang, Liwei Song, Zhi Li, Zhenya Huang, Enhong Chen. Towards Personalized Evaluation of Large Language Models with An Anonymous Crowd-Sourcing Platform. (WWW 2024).

[53] Junchen Fu, Fajie Yuan, Yu Song, Zheng Yuan, **Mingyue Cheng**, et al. Exploring Adapter-based Transfer Learning for Recommender Systems. *ACM WSDM 2024*: 208–217, Mar 2024. [Code]

2023 年及以前

[54] Zhiding Liu, **Mingyue Cheng**, Zhi Li, Zhenya Huang, Qi Liu, Yanhu Xie, Enhong Chen. Adaptive Normalization for Non-stationary Time Series Forecasting: A Temporal Slice Perspective. *NeurIPS 2023*: 36, New Orleans, 2023. [Code]

[55] 耿杰, 刘春丽*, 魏雪梅, **程明月**, 袁昆, 李洋, 刘业政. 基于用户重购行为的产品推荐方法. 计算机研究与发展 2023, 60(8): 1795–1807.

[56] **Mingyue Cheng**, Qi Liu*, Zhiding Liu, Zhi Li, Yucong Luo, Enhong Chen. FormerTime: Hierarchical Multi-scale Representation for Multivariate Time Series Classification. *ACM WWW 2023*: 1437–1445, Austin, 2023. [Code]

[57] Wenqiang He, **Mingyue Cheng**, Qi Liu*, Zhi Li. ShapeWordNet: An Interpretable Shapelet Neural Network for Physiological Signal Classification. *DASFAA 2023*: 353–369, Tianjin, 2023.

[58] **Mingyue Cheng**, Zhiding Liu+, Qi Liu*, Shenyang Ge, Enhong Chen. Towards Automatic Designing of Deep Hybrid Network Architecture for Sequential Recommendation. *ACM WWW 2022*: 1923–1932. [Code]

[59] Zhiding Liu, **Mingyue Cheng**, Zhi Li, Qi Liu, Enhong Chen*. One Person, One Model — Learning Compound Router for Sequential Recommendation. *IEEE ICDM 2022*. [Code]

[60] Junzhe Jiang, **Mingyue Cheng**, Qi Liu*, Zhi Li, Enhong Chen. Nested Named Entity Recognition from Medical Texts: A Multi-task Learning Approach. *CAAI CICAI 2022*: 248–259.

[61] Runlong Yu, Qi Liu*, Yuyang Ye, **Mingyue Cheng**, Enhong Chen, Jianhui Ma. Collaborative List-and-Pairwise Filtering from Implicit Feedback. *IEEE Transactions on Knowledge and Data Engineering (IEEE TKDE)*, 2022, 34(6): 2667–2680.

[62] Kai Zhang, Qi Liu*, Zhenya Huang, **Mingyue Cheng**, Kun Zhang, Mengdi Zhang, Wei Wu, Enhong Chen. Graph Adaptive Semantic Transfer for Cross-domain Sentiment Classification. *ACM SIGIR 2022*: 1566–1576.

[63] **Mingyue Cheng**, Fajie Yuan+, Qi Liu*, Xin Xin, Enhong Chen. Learning Transferrable User Representations with Sequential Behaviors via Contrastive Pre-training. *IEEE ICDM 2021*: 51–60.

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[65] Linan Yue, Qi Liu*, Han Wu, Kai Zhang, Yanqing An, **Mingyue Cheng**, Biao Yin, Dayong Wu. NeurJudge: A Circumstance-aware Neural Framework for Legal Judgment Prediction. *ACM SIGIR 2021*: 973–982.

[66] Yanqing An, Qi Liu*, Han Wu, Kai Zhang, Linan Yue, **Mingyue Cheng**, Hongke Zhao, Senchao Yuan, Enhong Chen. LawyerPAN: A Proficiency Assessment Network for Trial Lawyers. *ACM SIGKDD 2021*: 5–13.

[67] **Mingyue Cheng**, Runlong Yu, Qi Liu*, Hongke Zhao, Hefu Zhang, Enhong Chen. Alpha-Beta Sampling for Pairwise Ranking in One-Class Collaborative Filtering. *IEEE ICDM 2019*: 1000–1005.

[68] **程明月**, 刘淇*, 李徵, 于润龙, 高维博, 陈恩红. 多重对级贝叶斯个性化排序算法. (南京信息工程大学学报自然科学版, 2019(03): 302–308).

预印本

[69] Zhiding Liu+, Jiqian Yang+, **Mingyue Cheng***, Zhi Li*, Enhong Chen*. World2TS: Real-World Event Context for Time-Series Forecasting. (预印本). [Website]

[70] **Mingyue Cheng**, Xiaoyu Tao, Qi Liu, Ze Guo, Enhong Chen. Position: Beyond Model-Centric Prediction — Agentic Time Series Forecasting. (预印本).

[71] Xiaoyu Tao, **Mingyue Cheng**, Chuang Jiang, Tian Gao, Huanjian Zhang, Yaguo Liu, Qi Liu. Cast-R1: Learning Tool-Augmented Sequential Decision Policies for Time Series Forecasting. (预印本). [Code]

[72] Xiaoyu Tao, **Mingyue Cheng**, Bokai Pan, Chuang Jiang, Huanjian Zhang, Tian Gao, Yaguo Liu, Qi Liu, Enhong Chen. CastFSR: A Fast–Slow–Reflect Agentic Reasoning Framework for Context-Aware Time Series Forecasting. (预印本).

[73] Ze Guo, **Mingyue Cheng**, Xiaoyu Tao, Shuo Yu, Shangtong Ye. NeoResearch: Literature- and Experience-Guided Autonomous Time Series Model Discovery. (under review).

[74] Bokai Pan, Ruijie Yang, **Mingyue Cheng**, Jiahao Wang, Xiaoyu Tao. CastWorld: Benchmarking Context-Aware Agentic Time Series Forecasting Beyond Predictive Accuracy. (under review).

[75] Bokai Pan, **Mingyue Cheng***, Zhiding Liu, Shuo Yu, Xiaoyu Tao, Yuchong Wu, Qi Liu, Defu Lian, Enhong Chen. CastFlow: Learning Role-Specialized Agentic Workflows for Time Series Forecasting. (预印本). [Code]

[76] Daoyu Wang, Qingchuan Li, **Mingyue Cheng**, Jie Ouyang, Shuo Yu, Qi Liu, Enhong Chen. StepPO: Step-Aligned Policy Optimization for Agentic Reinforcement Learning. (预印本).

[77] **Mingyue Cheng**, Daoyu Wang, Shuo Yu, Qingchuan Li, Jie Ouyang, Yucong Luo, Yiju Zhang, Qi Liu*, Enhong Chen. A Comprehensive Survey of the LLM-Based Agent: The Contextual Cognition Perspective. (预印本).

科研项目

中国科学院基础与交叉前沿科研先导专项（B 类） 2026.07 – 2029.06

大模型自主交互学习机制及方法。

承担角色： 项目负责人。

新一代人工智能国家科技重大专项 2026.08 – 2028.07

科学数据治理工具链与数据集—化学领域。

承担角色： 项目骨干。

国家自然科学基金青年科学基金 C 类 2026.01 – 2028.12

跨域情境感知的时间序列表征学习及预测方法。

承担角色： 项目负责人。

中国科学技术大学青年创新基金项目 2027.01 – 2028.12

大模型智能体多轮交互学习与持续进化方法研究及应用。

承担角色： 项目负责人；计划于 2027 年 1 月开始。

中国科学技术大学新医学联合基金培育项目 2025.01 – 2026.12

双一流学科建设专项；基于围术期生理数据的时序建模方法及应用研究。

承担角色： 项目负责人。

安徽省自然科学基金 2024.09 – 2026.08

面向科技文献的表格语义理解与推理研究。

承担角色： 项目负责人。

企业科研支持：科大讯飞、华为、腾讯与快手。

开源项目与研究系统

- **Agent-R1**: 统一、模块化的智能体强化学习框架，支持多轮交互训练与算法比较。
- **Claw-R1**: 面向智能体强化学习的数据基础设施，支持交互轨迹采集、评估与治理。
- **CastClaw（星辰）**：面向时序预测研究的人机协同智能体框架，在关键节点引入人类确认。
- **CastMind（星思）**：融合历史时序、外部情境与领域知识，开展情境推理驱动的时序推演、预测修正与机制解释。代码尚未发布。
- **PaperScout（科言乐问）**：面向复杂科研查询的自主文献检索智能体，结合过程感知策略优化。

- **Academic Search**: 多源学术检索与元数据工具，支持记录校验、去重、引用追踪与 BibTeX 导出。
- **科言系列应用：**科言乐问，科学文献智能获取；科言文修，学术写作润色、校对与翻译。

学术服务

会议程序委员会与审稿

KDD（2023–2027）；**TheWebConf（2024–2027；2027 年 Area Chair）**；WSDM（2025、2027）；CIKM（2024、2025）；ICDM；**ICML（2025、2026；2027 年 Area Chair）**；**ICLR（2025、2027；2027 年 Area Chair）**；NeurIPS（2024–2026）；ACL（2025、2026）；EMNLP；AAAI（2026、2027）；SIGSPATIAL（2026）；IJCAI（2024–2026，2026 年 SPC）；SDM（2024）；DASFAA（2024、2025）。

2027 年条目为即将开展的任职。

期刊审稿

IEEE TKDE、TPAMI、TNNLS、TII、Transactions on Audio, Speech and Language Processing；ACM TKDD、TIST；Transactions on Machine Learning Research；Neurocomputing；Frontiers of Computer Science；Science China Information Sciences；《软件学报》；《计算机学报》。

学术组织任职

- 中国计算机学会人工智能与模式识别专业委员会，**执行委员**。
- 中国中文信息学会信息检索专业委员会，**通讯委员**。
- IEEE Task Force on Data-Efficient Agentic Learning（DEAL）。
- IEEE Task Force on AI for Time Series and Spatio-Temporal Data。

会议组织与学术讲习

Publication Chair，IEEE ICEBE 2026。

Program Chair，FedKDD/FedMAS 2026，KDD 2026 同期 Workshop。

Tutorial 组织者与报告人，*Context-Aware Time Series Forecasting: From Pattern Extrapolation to Cognitive Reasoning*，IEEE ICDM 2026 Industry Tutorial Session（已接收，待举行）。

代表性荣誉与奖励

- **2025 年**，中国科学技术大学红专青年人才。
- **2025 年**，Best of WSDM（优秀论文奖，*InstrucTime*）。
- **2024 年**，昇腾人工智能创新大赛全国总决赛金奖（指导教师）。
- **2024 年**，KDD Cup 2024，Task 2 与 Task 3 第二名。
- **2023 年**，中国国际大学生创新大赛国家级银奖（教育部）。
- **2023 年**，中国科学技术大学优秀毕业生。
- **2021 年**，博士研究生国家奖学金（教育部）。

教学

数据分析及实践，本科生课程，2024、2025、2026 年春季学期。

系统程序设计基础，本科生课程，2026 年春季学期。